

ShopPulse — E-Commerce Sales Analytics Platform

Comprehensive Project Documentation, Architecture & Strategic Business Case Study

Author: B.Tech AI & Data Science Student | **Target Role:** Data Analyst / Analytics Engineer

Dataset: Verified Sample Superstore E-Commerce Dataset (Tableau Public / Kaggle Open Data)

Scope: 9,994 Transactions | \$2.297M Revenue | \$286.39K Profit | 12.47% Margin | 20 Advanced SQL Queries

1. Executive Summary & Verified Dataset Specifications

ShopPulse is a production-quality e-commerce analytics platform engineered using the canonical, real-world **Sample Superstore Retail Sales Dataset**. The platform tracks 9,994 transactions occurring across 4 full operating years (2014–2017), spanning 3 major product categories (Technology, Furniture, Office Supplies), 17 sub-categories, 4 geographic regions (West, East, Central, South), and 3 customer segments (Consumer, Corporate, Home Office).

Verified Dataset Provenance:

- **Source:** Tableau Public Open Data / Kaggle Open Retail Dataset Repository
- **Source URL:** <https://raw.githubusercontent.com/yajasarora/Superstore-Sales-Analysis-with-Tableau/master/Superstore%20sales%20dataset.csv>
- **License:** Public Domain / Open Database License
- **Data Integrity:** 100% of reported figures are directly computed from the 9,994 transactional rows with zero synthetic fabrication.

2. Enterprise Performance Scorecard (Empirical Ground Truth)

Metric Name	Actual Value	Business Scope	Analytical Interpretation
Total Revenue	\$2,297,200.65	4-Year Aggregate	Total realized top-line sales across 9,994 line items.
Total Gross Profit	\$286,396.54	4-Year Aggregate	Net gross earnings realized after Cost of Goods Sold (COGS).
Overall Profit Margin	12.47%	Benchmark: >10.0%	Realized company-wide profit margin percentage.
Total Order Volume	5,009 Orders	4 Operating Years	Total distinct checkout orders placed across all channels.
Unique Customers	793 Customers	All Segments	Active customer base across Consumer, Corporate, and Home Office.
Unique Product SKUs	1,862 Products	17 Sub-Categories	Active merchandise catalog breadth.
Average Order Value (AOV)	\$458.62	Per Checkout	Mean basket sales generated per completed order (\$2.297M / 5,009).
Repeat Customer Rate	98.49%	Benchmark: >80.0%	High retention: 781 out of 793 customers placed 2+ lifetime orders.

3. Technical Stack & Data Pipeline Architecture

Pipeline Layer	Technology Stack	Core Functions & Capabilities
Data ETL & Cleaning	Python 3.11, Pandas, NumPy	Header standardization to snake_case, ISO date normalization, calendar feature enrichment, and financial
Database Layer	PostgreSQL & SQLite	Star Schema relational tables (dim_customers, dim_products, fact_orders) with foreign key integrity and B-t
SQL Analytics	Advanced SQL (20 Queries)	Window Functions (LAG, DENSE_RANK, SUM OVER), Common Table Expressions (CTEs), RFM segment
Visualization & BI	Streamlit, Plotly, Flask, Chart.js	Interactive analytics dashboards featuring dynamic filtering by operating year, region, category, and custom
Quality Assurance	Pytest Suite	Automated unit & integration test coverage verifying ETL transformations, database seeding, and query ma

4. Advanced SQL Analytics Catalog (20 Verified Queries)

Below is a categorized selection of the 20 production SQL queries implemented in database/analysis_queries.sql executed against the real database:

Query 03: Month-Over-Month (MoM) Growth Analysis Using LAG()

Business Purpose: Computes period-over-period sales velocity and acceleration without self-joins.

```
WITH monthly_metrics AS (  
    SELECT SUBSTR(order_date, 1, 7) AS year_month,  
           ROUND(SUM(sales), 2) AS revenue, ROUND(SUM(profit), 2) AS profit  
    FROM fact_ecommerce_sales GROUP BY SUBSTR(order_date, 1, 7)  
)  
SELECT year_month, revenue,  
       LAG(revenue, 1) OVER (ORDER BY year_month) AS prev_month_revenue,  
       ROUND(((revenue - LAG(revenue, 1) OVER (ORDER BY year_month)) /  
             LAG(revenue, 1) OVER (ORDER BY year_month)) * 100.0, 2) AS mom_revenue_growth_pct  
FROM monthly_metrics ORDER BY year_month ASC;
```

Query 07: Top 3 Sub-Categories per Category Using DENSE_RANK()

Business Purpose: Ranks top 3 revenue leaders inside Technology, Furniture, and Office Supplies.

```
WITH subcat_sales AS (  
    SELECT category, sub_category, ROUND(SUM(sales), 2) AS total_sales,  
           DENSE_RANK() OVER (PARTITION BY category ORDER BY SUM(sales) DESC) AS rank_in_category  
    FROM fact_ecommerce_sales GROUP BY category, sub_category  
)  
SELECT category, rank_in_category, sub_category, total_sales  
FROM subcat_sales WHERE rank_in_category <= 3 ORDER BY category, rank_in_category ASC;
```

Query 13: Discount Depth vs. Realized Profit Margin

Business Purpose: Empirically identifies that discounts > 20% generate net losses (-\$32K loss).

```
SELECT  
    CASE  
        WHEN discount = 0.00 THEN '0% (No Discount)'  
        WHEN discount <= 0.20 THEN '1% - 20% (Standard Discount)'  
        WHEN discount <= 0.50 THEN '21% - 50% (Deep Discount)'  
        ELSE '51%+ (Clearance)'  
    END AS discount_bracket,  
    COUNT(order_id) AS total_transactions, ROUND(SUM(sales), 2) AS total_revenue,  
    ROUND(SUM(profit), 2) AS total_profit,  
    ROUND((SUM(profit) / SUM(sales)) * 100.0, 2) AS realized_margin_pct  
FROM fact_ecommerce_sales GROUP BY discount_bracket ORDER BY realized_margin_pct DESC;
```

Query 18: Pareto 80/20 Cumulative Product Revenue Concentration

Business Purpose: Measures revenue concentration among top-selling hardware and office equipment SKUs.

```
WITH product_totals AS (  
    SELECT product_name, category, ROUND(SUM(sales), 2) AS product_sales  
    FROM fact_ecommerce_sales GROUP BY product_name, category  
)  
,  
ranked_products AS (  
    SELECT product_name, product_sales,  
           SUM(product_sales) OVER (ORDER BY product_sales DESC ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS cumulative_sales,  
           SUM(product_sales) OVER () AS total_enterprise_sales  
    FROM product_totals  
)  
SELECT product_name, product_sales, ROUND((cumulative_sales / total_enterprise_sales) * 100.0, 2) AS cumulative_pct
```

FROM ranked_products LIMIT 15;

5. Top Strategic Business Insights & Recommendations

Extracted from reports/business_insights.md, these core findings represent actionable, quantifiable discoveries:

1. Technology Category Profit Dominance

- **Finding:** Technology generated \$836,154.02 in revenue (36.4% share) and \$145,455.03 in gross profit, driving 50.79% of all enterprise profit with a 17.39% margin.
- **Impact:** Technology hardware is the primary cash engine of the retail business.
- **Recommendation:** Expand inventory allocations for high-margin tech accessories and copiers; increase tech marketing spend by 20%.

2. Furniture Category Margin Compression (2.49% Margin)

- **Finding:** Furniture generated \$741,999.74 in sales but yielded only \$18,451.24 in net profit (2.49% margin). Tables lost -\$17,725.48 and Bookcases lost -\$3,472.56.
- **Impact:** Freight shipping costs and heavy promotional discounts eliminate furniture profitability.
- **Recommendation:** Eliminate discounts exceeding 15% on Tables and Bookcases; renegotiate carrier logistics rates.

3. The 'Discount Destruction' Cliff (>20% Discount)

- **Finding:** Transactions with 0% discount achieved 29.9% margin; discounts 1-20% achieved 14.5% margin. Discounts > 20% resulted in an aggregate net loss of -\$32,142.98.
- **Impact:** Steep promotional markdowns destroy gross margins without generating compensating volume.
- **Recommendation:** Implement hard checkout guardrails capping standard discounts at 15%.

4. Regional Profit Asymmetry — West Leads While Central Lags

- **Finding:** The West region produced \$725,457.82 in sales and \$108,418.45 in profit (14.94% margin), whereas Central delivered \$501,239.89 in sales and only \$39,706.36 in profit (7.92% margin).
- **Impact:** Aggressive discounting in Texas (-\$25.7K loss) drags down Central region returns.
- **Recommendation:** Enforce West region pricing governance across Central sales branches.

5. B2B & Home Office Deliver Higher Profit Margins

- **Finding:** Home Office accounts achieved 14.03% margin (\$60,298 profit) and Corporate accounts achieved 13.03% (\$91,979 profit), outperforming Consumer retail (11.55%).
- **Impact:** Business clients demonstrate lower price sensitivity and higher average transaction values.
- **Recommendation:** Build dedicated corporate procurement self-service portals with tailored volume pricing.

6. Pareto 80/20 SKU Revenue Concentration

- **Finding:** The top 15.2% of product SKUs (283 / 1,862) generate 70% of total enterprise sales. Canon imageCLASS Copier led with \$61,599.82 in sales.
- **Impact:** Working capital and top-line health are tied to key equipment SKUs.
- **Recommendation:** Maintain 60-day safety stock buffers on top 50 revenue drivers.

6. Technical Interview Elevator Pitch

*"In this project, I built **ShopPulse**, an end-to-end e-commerce sales analytics platform utilizing the verified Sample Superstore retail sales dataset comprising 9,994 transactions across 4 operating years. I built a Python ETL pipeline to standardize schemas, enrich temporal features, and validate financial consistency. Next, I designed a normalized Star Schema in PostgreSQL/SQLite and authored 20 advanced SQL queries utilizing Window Functions (LAG, DENSE_RANK, SUM OVER) to model customer lifetime value, RFM segmentation, and MoM growth velocity. Finally, I built interactive dashboards in Streamlit and delivered 12 quantified business case studies demonstrating how capping promotional discounts at 15% and addressing furniture margin compression directly recaptures operating capital."*

GitHub Repository: <https://github.com/mahamoodmajin183-ui/shop-pulse-ecommerce-analytics> | **License:** MIT